

93
100

Good!

Prob 1(10) Show that the polar equation $r = a \sin \theta$ represents a circle where $a \neq 0$, and find its center and radius.

1) $x = r \cos \theta$ $x^2 + y^2 = r^2$
 $y = r \sin \theta$ $\tan \theta = \frac{y}{x}$

$r = a \sin \theta$

$r^2 = a \sin \theta \cdot r \rightarrow y = r \sin \theta$

$\therefore x^2 + y^2 = ay \leftarrow \text{conversion to cartesian}$
 $x^2 + y^2 - ay = 0$ $\leftarrow \text{Completing the square}$
 $x^2 + y^2 - ay + \frac{a^2}{4} = 0 + \frac{a^2}{4}$

$x^2 + (y - \frac{a}{2})^2 = \frac{a^2}{4}$

$(x - h)^2 + (y - k)^2 = r^2$
 Equation of circle

$\therefore h = 0$

$k = \frac{a}{2}$

$r^2 = \frac{a^2}{4}$ $r = \sqrt{\frac{a^2}{4}} = \frac{a}{2}$ $\leftarrow \text{radius}$

$\therefore \text{Center} = (\frac{a}{2}, 0) \leftrightarrow (x, y) = (0, \frac{a}{2})$

Prob 2(20) Sketch the curve for $r = \frac{1}{\theta}$ for $\frac{\pi}{2} \leq \theta \leq 2\pi$. Also, find the length of the curve.
 (Hint) You may use the following formula.

$\int \frac{1}{\sqrt{t^2 + 1}} dt = \sinh^{-1} t + C$

2) $r = \frac{1}{\theta}$ for $\frac{\pi}{2} \leq \theta \leq 2\pi$

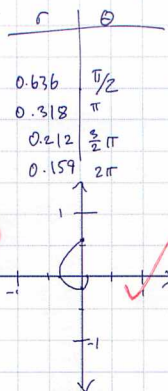
$x = r \cos \theta \rightarrow x = \frac{1}{\theta} \cos \theta$
 $y = r \sin \theta \rightarrow y = \frac{1}{\theta} \sin \theta$

curve length formula = $L = \int_a^b \sqrt{r^2 + (\frac{dr}{d\theta})^2} d\theta$

$= \int_{\pi/2}^{2\pi} \sqrt{\frac{1}{\theta^2} + 1} d\theta$

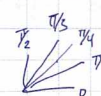
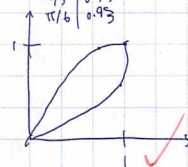
$= \left[\ln(\sqrt{x^2 + 1} + x) - \frac{\sqrt{x^2 + 1}}{x} \right]_{\pi/2}^{2\pi}$
 $= (2.5993 - 1.0126) - (1.1334 - 1.1854)$

$L = 1.4767$



Prob 3(10) Sketch the curve for $r = \sqrt{\sin(2\theta)}$ for $0 \leq \theta \leq \frac{\pi}{2}$ and find the area enclosed by the curve.

3) $\theta \mid r$
 $0 \mid 0$
 $\pi/4 \mid 1$
 $\pi/2 \mid 0$
 $\pi/3 \mid 0.93$
 $\pi/6 \mid 0.93$



Formula for Area = $\int_a^b \frac{1}{2} r^2 d\theta$

$A = \int_0^{\pi/2} \frac{1}{2} \sin 2\theta d\theta$

$A = \left[-\frac{1}{4} \cos 2\theta \right]_0^{\pi/2}$
 $= -\frac{1}{4} \cos 2(\frac{\pi}{2}) - (-\frac{1}{4} \cos 2(0))$
 $= \frac{1}{4} - (-\frac{1}{4}) = \frac{1}{4} + \frac{1}{4} = \frac{2}{4} = \frac{1}{2}$

$\therefore A = \frac{1}{2}$
 Area enclosed by the curve

Prob 4(10) Let us consider a curve whose vector equation is given by

$r(t) = e^{-t} \cos t i + e^{-t} \sin t j + a k$ where a is constant. Using the fact that an infinitesimal length of the curve is expressed by $ds = \sqrt{u^2 + v^2} dt$, find the length of the curve for $0 \leq t \leq 2\pi$.

a) $r(t) = e^{-t} \cos t i + e^{-t} \sin t j + a k$

$0 \leq t \leq 2\pi$

$ds = \sqrt{u^2 + v^2} dt$

$s(t) = \int_a^b \sqrt{(f'(t))^2 + (g'(t))^2 + (h'(t))^2} dt$

$f'(t) = -e^{-t} \sin t - e^{-t} \cos t$ $g'(t) = e^{-t} \sin t + e^{-t} \cos t$ $h'(t) = 0$

$= \int_0^{2\pi} \sqrt{(-e^{-t} \sin t - e^{-t} \cos t)^2 + (e^{-t} \sin t + e^{-t} \cos t)^2} dt$

$= \int_0^{2\pi} 2(e^{-t} \cos t)^2 + 2(e^{-t} \sin t)^2 dt = \int_0^{2\pi} 2e^{-2t} dt$

$= \int_0^{2\pi} \sqrt{2} \cdot e^{-t} dt = \sqrt{2} [-e^{-t}]_0^{2\pi}$

Prob 5(20) Find the smallest radius of curvature of the function $y = \exp(x)$.

$$y = e^x$$

5)

$$y = e^x$$

$$\frac{dy}{dx} = e^x$$

$$\frac{d^2y}{dx^2} = e^x$$

$$R = \frac{(1 + (\frac{dy}{dx})^2)^{3/2}}{|\frac{d^2y}{dx^2}|}$$

$$= \frac{(1 + e^{2x})^{3/2}}{|e^x|}$$

$$R'(x) = \frac{\frac{3}{2}(1+e^{2x})^{1/2}(2e^{2x}) \cdot e^x - e^x \cdot (1+e^{2x})^{3/2}}{e^{2x}} = 0$$

$$\frac{3}{2}(2e^{2x}) - (1+e^{2x}) = 0$$

$$3e^{2x} - 1 - e^{2x} = 0$$

$$2e^{2x} - 1 = 0$$

$$2e^{2x} = 1$$

$$e^{2x} = \frac{1}{2}$$

$$e^x = \sqrt{\frac{1}{2}} = \frac{\sqrt{2}}{2} \text{ we're going to accept the answer of } \frac{\sqrt{2}}{2}$$

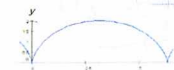
$$\therefore x = \ln \frac{\sqrt{2}}{2} = -0.3465 \approx -0.347$$

$$\therefore R(\ln \frac{\sqrt{2}}{2}) = \frac{(1 + \frac{1}{2})^{3/2}}{\frac{1}{\sqrt{2}}} = \boxed{2.598}$$

The smallest radius is 2.598 units

Prob.6(30) A curve of cycloid (right figure) is expressed by the following parametric equations:

$$x = a(t - \sin t), \quad y = a(1 - \cos t)$$



where t ($0 \leq t \leq 2\pi$) is a parameter.

(a) Show that the unit tangent vector $\mathbf{T}(t)$ to the cycloid is expressed by

$$\mathbf{T}(t) = \sin\left(\frac{t}{2}\right) \mathbf{i} + \cos\left(\frac{t}{2}\right) \mathbf{j}$$

(b) Find the normal vector $\mathbf{N}(t)$ and binormal vector $\mathbf{B}(t)$ on the cycloid.

(c) Find the arc-length $s(t)$ of the cycloid from the origin as a function of t .

6)

$$x = a(t - \sin t) \quad y = a(1 - \cos t) \quad t \in [0, 2\pi]$$

$$a) \frac{d\mathbf{r}}{dt} = a(1 - \cos t)\mathbf{i} + a \sin t \mathbf{j}$$

$$\left| \frac{d\mathbf{r}}{dt} \right| = 2a \sin \frac{t}{2}$$

$$10 \quad \mathbf{T}(t) = \frac{\frac{d\mathbf{r}}{dt}}{\left| \frac{d\mathbf{r}}{dt} \right|}$$

$$= \frac{1 - \cos t}{2 \sin \frac{t}{2}} \mathbf{i} + \frac{\sin t}{2 \sin \frac{t}{2}} \mathbf{j}$$

$$= \sin \frac{t}{2} \mathbf{i} + \cos \frac{t}{2} \mathbf{j} = \mathbf{T}(t) = \sin\left(\frac{t}{2}\right) \mathbf{i} + \cos\left(\frac{t}{2}\right) \mathbf{j}$$

b) find $\mathbf{N}(t)$ and $\mathbf{B}(t)$

$$\mathbf{B}(t) = \mathbf{T}(t) \times \mathbf{N}(t)$$

10

$$\frac{d\mathbf{T}}{dt} = \frac{1}{2} \cos \frac{t}{2} \mathbf{i} - \frac{1}{2} \sin \frac{t}{2} \mathbf{j}$$

$$\left| \frac{d\mathbf{T}}{dt} \right| = \frac{1}{2}$$

$$\therefore \mathbf{N}(t) = \frac{\frac{d\mathbf{T}}{dt}}{\left| \frac{d\mathbf{T}}{dt} \right|} = \frac{\frac{1}{2} \cos \frac{t}{2} \mathbf{i} - \frac{1}{2} \sin \frac{t}{2} \mathbf{j}}{\frac{1}{2}}$$

$$= \cos \frac{t}{2} \mathbf{i} - \sin \frac{t}{2} \mathbf{j}$$

$$\mathbf{B}(t) = \mathbf{T} \times \mathbf{N} = \left(\sin \frac{t}{2} \mathbf{i} + \cos \frac{t}{2} \mathbf{j} \right) \times \left(\cos \frac{t}{2} \mathbf{i} - \sin \frac{t}{2} \mathbf{j} \right)$$

$$\mathbf{B}(t) = -\sin^2 \frac{t}{2} \mathbf{k} + \cos^2 \frac{t}{2} (-\mathbf{k}) = -\mathbf{k}$$

(c) Find the arc-length $s(t)$ of the cycloid from the origin as a function of t .

g)

$$x = a(t - \sin t) \quad y = a(1 - \cos t) \quad 0 \leq t \leq 2\pi$$

$$5 \quad L = \int_0^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

$$\frac{dx}{dt} = a(1 - \cos(t)) \quad \frac{dy}{dt} = a \sin t$$

$$L = \int_0^{2\pi} \sqrt{a^2(1 - \cos(t))^2 + a^2 \sin^2 t} dt$$

$$= \int_0^{2\pi} \sqrt{a^2(1 - \cos^2(t) + \sin^2 t)} dt$$

$$5 = \int_0^{2\pi} \sqrt{a^2(1 - \cos^2(t) + \sin^2 t)} dt$$

$$= \int_0^{2\pi} a \sqrt{\sin^2(x) - \cos^2(x) + 1} dx$$

$$= \int \sqrt{2} |a| \sin(x) dx$$

$$= \sqrt{2} |a| \int \sin(x) dx = \sqrt{2} |a| (-\cos(x)) = \left[\sqrt{2} |a| \cos(x) + C \right]_0^{2\pi}$$

$$\sqrt{2} |a| \cos\left(\frac{2\pi}{2}\right) + C - \sqrt{2} |a| \cos(0) + C$$

$$= \sqrt{2} |a| \cos(\pi) + C - \sqrt{2} |a| (1) + C$$

$$\text{Arc length} \rightarrow s = -\sqrt{2} |a| + 2C$$